Parking Management System

A C++ implementation for vehicle parking allocation and ticketing

System Architecture Overview

Three core classes work together to manage parking operations

- Vehicle: Tracks owner and vehicle details
- ParkingSlot: Manages slot availability
- Ticket: Handles timing and fees

The Vehicle Class

Attributes

Vehicle number, type, owner name, contact number

Setters

Update any vehicle detail dynamically

Getters

Retrieve stored vehicle information

ParkingSlot Class Operations

Park Vehicle

Check slot availability, mark occupied

Verify Status

Return occupancy state

Remove Vehicle

Free slot, reset tracking

Ticket System & Fee Calculation

Time Tracking

Automatically records entry and exit timestamps

Fee Structure

- First hour: $20
- Additional hours: $10 per hour

Fee Calculation Logic

01

Calculate Duration

Compute hours between entry and exit

02

Apply Base Rate

Set initial $20 fee for first hour

03

Add Overtime

Charge $10 for each additional hour

The ParkingLot Manager

Find Free Slot

Linear search through available parking spaces

Issue Ticket

Create unique ticket ID and start timer

Track Tickets

Maintain vector of active and expired tickets

Key Operations

1

Park Vehicle

Allocate slot, generate ticket, display slot number and ticket ID

2

Exit Vehicle

Validate ticket ID, calculate fee, free slot

3

Display Available

Show all unoccupied slots

4

List Tickets

Show all issued tickets with details

Implementation Highlights

- ## Vector-Based Storage

  Vectors for slots and tickets provide dynamic sizing

- ## Linear Search

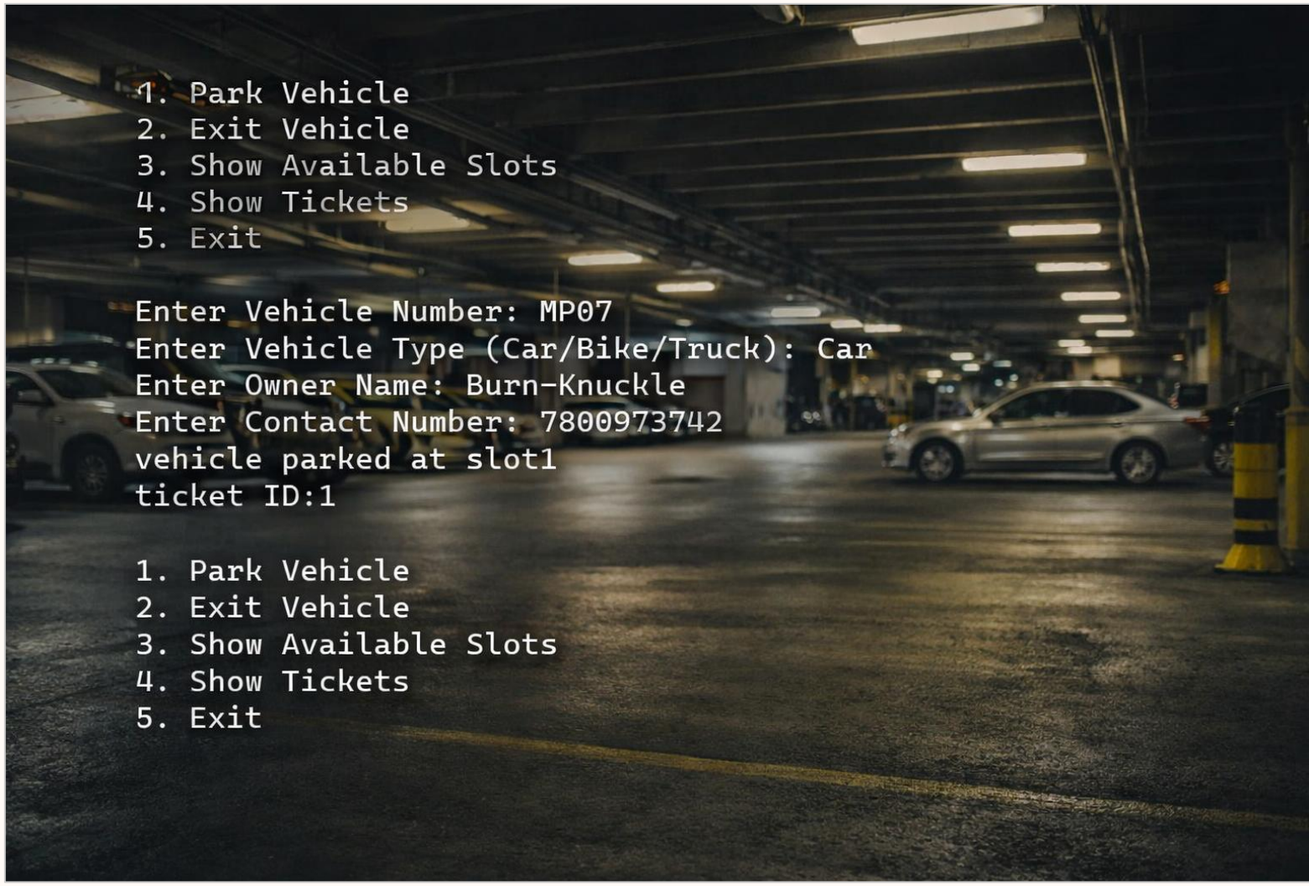
  Sequential slot finding (O(n) complexity)

- ## Ticket Validation

  Linear search through tickets ensures valid exit

Menu-driven interface with 5 options:

1. Park Vehicle - Enter details, allocate slot, issue ticket
2. Exit Vehicle - Enter ticket ID, calculate fee
3. Show Available Slots - List free slots
4. Show Tickets - View all issued tickets
5. Exit Program - Terminate system



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